

The Self-Prioritization Effect Database with Standardized Meta-Data for Experimental Task

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Abstract

The self-matching task, also known as self-tagging task, self-associative learning task, is widely used to investigate how self-related information is processed. In a typical self-matching task, participants first associate social labels (e.g., self, friend, and stranger) with geometric shapes (e.g., circle, triangle, and square) and then finish a matching task where they are asked to decide whether the shape-label pairings match the learned associations. Previous studies using this task have found faster and more accurate responses to self- than other-related pairings, i.e., the self-prioritization effect (SPE). Following the credibility revolution, a substantial number of these studies openly shared their data, created great potential for data re-using. However, there lacks of a standardized framework for sharing trial-level data generated by self-matching task, making it difficult to integrate existing open datasets. To fill the gap, we developed a three-level framework for organizing the metadata and trial-level data. We created machine-readable JSON files to document task implementations, including Physical Environment, Experimental Design, Block Structure, Trial Structure, and Stimulus Properties. We applied this framework to data available from 49 studies (89 experiment-level datasets, 4,875 participants, and over 2.1 million test trials). We demonstrated the value of the standardized open database through three examples: the effect of baseline conditions on SPE magnitude; the variability of SPE under nonmatch conditions; and the relationships between task details (duration of presenting time and number of trials) and SPE magnitude. The three examples show that standardizing meta-data across studies is what makes such aggregated analyses possible: the SPE was reliably positive against every baseline but differed reliably only between the baselines with the largest and the smallest point estimates; the SPE under nonmatching conditions was present but depended on how self-relevance was operationalized; and the magnitude of the SPE was related to stimulus presentation duration but not to the number of trials. Our standardized SPE database provides a valuable resource for the field. Moreover, our standardized framework for self-matching task is also valuable for developing similar databases for other cognitive tasks.

Keywords: Self-matching task, self-prioritization effect, open data, experimental design.

Introduction

The self plays a central role in cognition. Self-relevant information is processed faster and more accurately across perception, attention, and memory domains, which is known as self-prioritization effect (SPE) (Sui et al., 2012; Schäfer, Frings, et al., 2016; Constable et al., 2019). The self-matching task, also known as self-tagging task and self-associative learning task, has emerged as an important paradigm for investigating SPE. In a typical self-matching task, participant first associate geometric shapes with self- and non-self-relevant labels (e.g., "I", "Friend", and "Stranger") and then judge a shape-label pair presented on the screen match the learned association or not (Sui et al., 2012). Studies employed this task found better performance for self- than non-self-related stimuli (Liu et al., 2025; Schäfer, Wesslein, et al., 2016; Sui et al., 2012). Importantly, this effect has been observed across diverse cultural contexts (Constable et al., 2019; Grossmann & Jowhari, 2018) and different sensory modality (e.g., Schäfer, Wesslein, et al., 2016). This robust effect suggested a fundamental role of self-relevance in human cognition (Sui & Gu, 2017; Sui & Humphreys, 2015; Sui & Rotshtein, 2019).

In the era of credibility revolution, open data becomes a new norm (Houtkoop et al., 2018; Milham et al., 2018; Munafò et al., 2017). Specialized databases that aggregate data from multiple studies on particular phenomena provide unique opportunities for theory testing and discovery (Haaf et al., 2024; Rodriguez & Williams, 2022). Such resources allow researchers to examine effect sizes across diverse experimental conditions, identify moderating variables, and generate new hypotheses based on patterns that emerge only when data are aggregated across many studies (Cheung & Jak, 2016; Moreau & Gamble, 2022). The Attentional Control

Data Collection (ACDC) is a good example, which provides a structured open database for attentional control experiments that has facilitated meta-analyses and methodological advancements (Haaf et al., 2024). Similarly, the Confidence Database aggregates data from hundreds of experiments on confidence judgments and meta-cognitive performance into a unified repository (Rahnev et al., 2020). The Confidence Database enabled researchers to examine individual differences, compare computational models, and investigate theoretical questions that cannot be addressed using isolated datasets (Rahnev, 2025). While many SPE studies made their trial-level data publicly available, no standardized, large-scale dataset exists.

For cognitive paradigms such as the self-matching task, the lack of standardized meta-data and data organization slows down the reuse of open data scattered on the internet. For example, the primary outcome, reaction times, may be named differently across studies, such as RT, Response-Time, or Latency. Similarly, accuracy of each trial may use inconsistent coding schemes, such as 0 vs 1, True vs False, or Correct vs Error. The variable names of critical independent variables, such as the identities used in the self-matching task, may also be labelled differently across studies. More importantly, critical methodological information for implementing the task, such as the assignment of shapes to identities, stimulus presentation duration, trial structure, and participant exclusion criteria, is often absent from the raw data files and can only be found by reading the original papers. These issues substantially increase the effort required for reusing open data.

To address these issues, we developed a standardized framework for data organization and documentation, applied this framework to available open data, and formed the SPE database. To capture the methodological details systematically, we decomposed the self-matching task

into five components and documented information of each with standardized metadata files. We then collected all available datasets and manually standardized both behavioral variables and methodological descriptors to ensure cross-study comparability. In total, data from 49 studies, comprising 89 experiment-level datasets, were included. The SPE database enables analyses that would be impossible for small sample size studies.

Method

Folder Structure

Following the FAIR principle (Wilkinson et al., 2016), we adopted a three-level folder structure for the database: root folder, study folder, and experiment folder (if there are multiple experiments). Each level is accompanied by standardized metadata encoded in machine-readable JSON files, facilitating both human interpretation and computational accessibility.

The root folder contains information spanning all studies included in the database. The study sub-folder contains information related to a specific study, defined as one or more experiments reported within a single publication, preprint, or thesis. The experiment folders include individual experimental data, typically representing a group of participants completing a specific version of the self-matching task across one or multiple sessions and assessment time points.

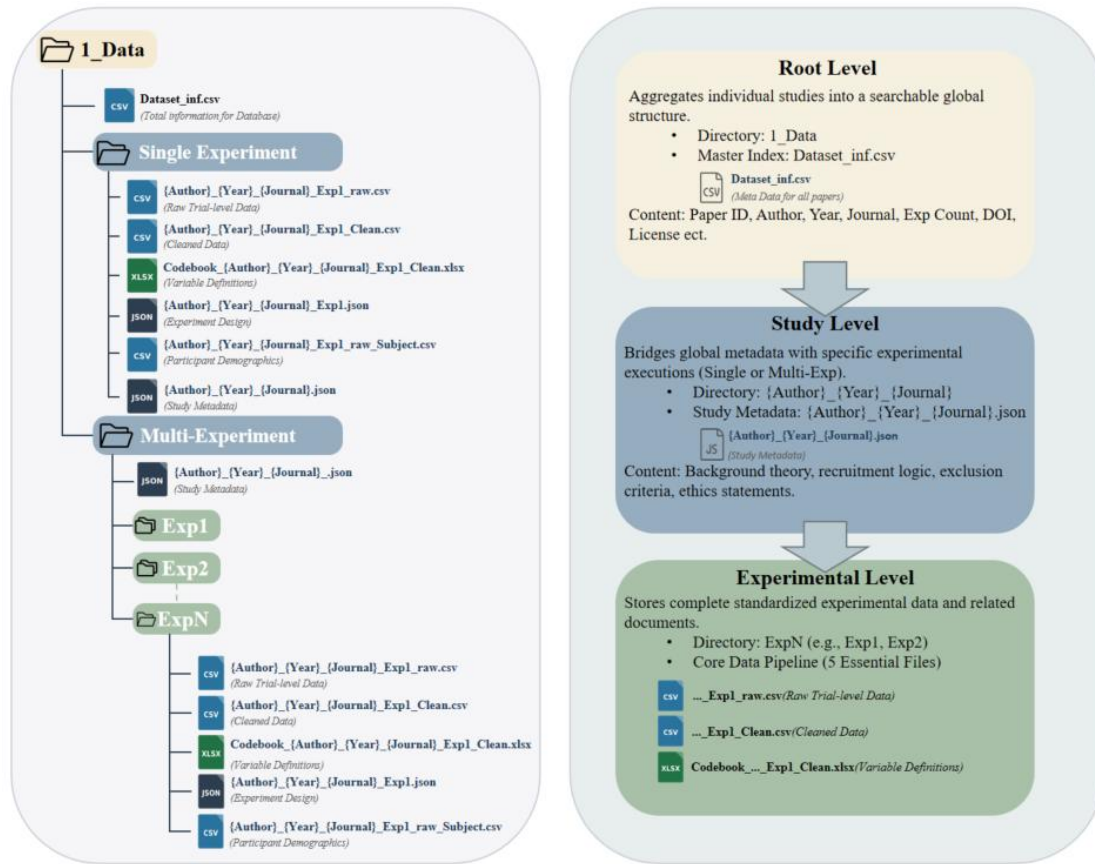


Figure 1. Database folder structure.

The root folder also contains a master metadata file ("Dataset_inf.csv"), which serves as the primary index for the entire database, and study-specific sub-folders (e.g., "Amodeo_2024_CABN"). Each row in this file represents one experiment-by-participant-group unit and is uniquely identified by the combination of the study folder ('Folder_Name'), the experiment identifier ('Exp'), and the participant group ('subj_Group'); the concatenation of these three fields forms the 'ID' column, which is the key across the database. Each column of "Dataset_inf.csv" corresponds to a metadata field of the studies, such as authors, publication year, journal name (where applicable), Digital Object Identifiers (DOIs), and sample descriptors. Importantly, the study-specific folder ('Folder_Name') was included in "Dataset_inf.csv" for clear mapping between studies and their data folders.

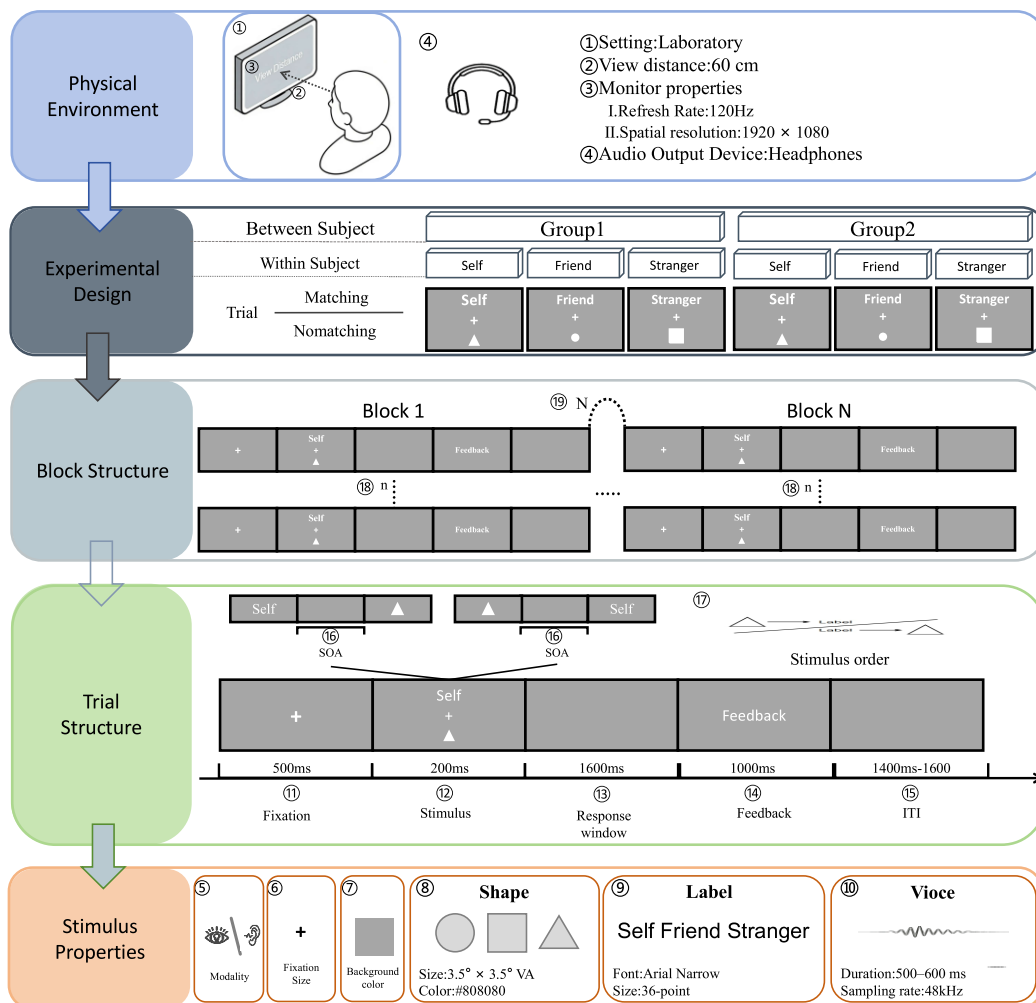
The study-specific folder contains a JSON file, e.g., 'Amodeo_2024_CABN.json', and

experiment-specific sub-folders. The JSON file contains study-level information such as the theoretical background, a summary of the study, its main conclusion, the publication year and journal (where applicable), the country and city in which the data were collected, and the Digital Object Identifier. The experiment-specific subfolders correspond to the individual experiments reported within the study. However, if a study contains only a single experiment, the experimental-level data are stored in the study-level folder directly and no experiment-specific subfolder was created.

For those studies reported more than one experiment, each experiment has its own subfolders (e.g., 'Exp1', 'Exp2'). Each experimental-level sub-folder include the following files: a CSV file for participants' demographic information (e.g., "Amodeo_2024_CABN_Exp1_subj_info.csv"), a JSON file for the experiment design (e.g., "Amodeo_2024_CABN_Exp1.json"), a CSV file for raw trial-level behavioral data file (e.g., "Amodeo_2024_CABN_Exp1_raw.csv"), a CSV file for the minimally preprocessed dataset (i.e., clean data, e.g., "Amodeo_2024_CABN_Exp1_Clean.csv"), and an xlsx file serves as the codebook for the clean data (e.g., "Codebook_Amodeo_2024_CABN_Exp1_Clean.xlsx"). To facilitate efficient data retrieval and readability, all file names include four pieces of information — last name of the first author, publication year, journal abbreviation, and experiment identifier. Additionally, an optional fifth element specifies the content or function of the file (e.g., "_subj_info" means information related to participants, "_raw" means unprocessed data from original studies, "Codebook_" marks the codebook of the clean data). Together, these rules enhance data interoperability and facilitate large-scale secondary analyses.

Standardization of the task implementation

To further facilitate cross-study integration of datasets, we developed a framework for standardizing the implementation of the self-matching task. This framework dissects the task into five components: Physical Environment, Experimental Design, Block Structure, Trial Structure, and Stimulus Properties (Figure 2). These five components zoom from the environment in which the experiment was done to the details of stimuli. Also, these details may also play a role in cognitive processes (e.g., Gold & Shadlen, 2007; Ratcliff et al., 2016), and allow nuanced cross-study analyses. We standardized all these five components of each experiment and stored them in the experiment-specific json file (e.g., "Amodeo_2024_CABN_Exp1.json") whenever relevant information is available.



```

Data > Amedeo_2024_CABN > {} Amedeo_2024_CABN_Exp1.json > ...
1  {
2    "exp1": {
3      "schema_version": "2",
4      "Collected_date": "/",
5      "Physical_Environment": {
6        "Location": "Ghent University (implied by ethical committee approval)",
7        "Setting": "Electrically shielded chamber (for EEG recording)",
8        "Equipment": {
9          "Presenting": "/",
10         "Monitor": "/",
11         "Software": "E-prime Studio, version 2.0.10.248"
12       },
13       "Viewing_distance": "/"
14     },
15     "Experimental_Design": {
16       "Conditions": "/"
17     },
18     "Block_Structure": {
19       "Practice_trials": "24 trials (repeatable up to 3 times if accuracy <70%)",
20       "Block_number": "6 blocks",
21       "Trial_number": "360 trials total (60 trials per block)"
22     },
23     "Trial_Structure": {
24       "Stimulus_order": "Simultaneously",
25       "SOA": "0 ms (simultaneous presentation)",
26       "Fixation_duration": "500 ms",
27       "Stimulus_duration": "Practice: 1000 ms; Test: 200 ms",
28       "Response_deadline": "Practice: 2400 ms; Test: 1600 ms",
29       "ITI": "Variable 1400-1600 ms",
30       "Feedback_duration": "1000 ms (practice phase only)"
31     },
32     "Stimulus_Properties": {
33       "Modality": "Visual",
34       "Fixation": "/",
35       "Shape": "3.7° × 3.7°",
36       "Label": "36-point Arial Narrow",
37       "Colors": {
38         "Stimulus": "White",
39         "Background": "Light gray"
40       }
41     },
42     "detail": ""
43   }
44 }

```

Figure 2. (a) Standardized implementation framework and five-component decomposition of the self-matching task; (b) an example of the experiment-specific json file.

The Physical Environment variables describe the overall environmental conditions under which participants completed the task, including basic settings (e.g., laboratory-based vs. online administration), conditions relevant to multimodal stimulus presentation. For visual stimuli, these encompass viewing distance, monitor refresh rate, display resolution, brightness (Lin et al., 2023). For auditory stimuli, these include hardware specifications (e.g., headphones vs. speakers), sound pressure levels, and ambient noise (Kirk & Cunningham, 2025). These

parameters determine how sensory information (e.g., visual or auditory signals) is physically presented to participants and they may modulate the sensory input of information processing (Bridges et al., 2020; Gold & Shadlen, 2007).

The Experimental Design variables define the structure of the study, including both between-subject factors (e.g., cultural group) and within-subject manipulations (e.g., Identity and Matching conditions). What distinguishes this level from the other levels is that design variables specify *what* is being manipulated and compared for hypothesis testing, rather than *how* the task is procedurally implemented, although these aspects may overlap. This distinction is consistent with established reporting guidelines that differentiate methodological details from inferential variables (Appelbaum et al., 2018).

The Block Structure describe the block structure of the task, including number of blocks as well and details within a block: trial sequence and composition. Block Structure captures how multiple trials are organized within a block and often may strongly influence the strategies adopted by participants (Bogacz et al., 2006) or result in serial dependency effect (Cicchini et al., 2024). Usually, trials in self-matching task are pseudo-randomized such that no more than three identical trial types are presented consecutively.

The Trial Structure zoom in each trial and describe how each trial is implemented, including fixation duration, stimulus exposure, stimulus onset asynchrony (SOA), response window, and inter-trial interval (ITI). These parameters govern the temporal dynamics of stimuli presentation and are closely related to cognitive mechanisms such as evidence accumulation and response preparation (Ratcliff et al., 2016).

Finally, the Stimulus Properties describe details of the stimuli, including their modality,

their visual/auditory properties, and attributes of labels. The defining boundary of this level is whether a piece of information specifies *what* information is presented rather than *how* it is physically delivered, with the latter belong to Trial Structure.

Summary of data included in SPE database

The Self-Matching Database is hosted on the Open Science Framework (OSF) website (XXX) and Science Data Bank (XXX). In the current release (September 2026), the database contains 49 studies, corresponding to 107 experiment-by-participant-group entries (89 experiment-level data files) and aggregating 4,875 participants across 2,131,108 test trials (See Table 1 for details).

Table 1*Overview of Studies Included in the SPE Database*

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
1	Amodeo_2024_CABN	L Amodeo et al. (2024)	Exp1	Belgium	Dutch	37	geometric shape	360	NA	Lab Experiment
2	Amodeo_2024_CABN	L Amodeo et al. (2024)	Exp1	Belgium	Dutch	30	geometric shape	360	NA	Lab Experiment
3	Atzeni_2026_PsychRes	Atzeni (2026)	Exp1	Italy	Italian	140	geometric shape	200 (per session; duplicated/extra runs: 3 T2 + 4 T3 participants)	NA	Online Experiment
4	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp1	Austria	German	19 (0/19)	geometric shape	360	NA	Lab Experiment
5	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp1	Austria	German	16 (0/16)	geometric shape	360	NA	Lab Experiment
6	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp1	Austria	German	22 (0/22)	geometric shape	360	NA	Lab Experiment
7	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp1	Austria	German	21 (0/21)	geometric shape	360	NA	Lab Experiment
8	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp2	Austria	German	35 (0/35)	geometric shape	360	NA	Lab Experiment
9	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp2	Austria	German	35 (0/35)	geometric shape	360	NA	Lab Experiment
10	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp2	Austria	German	33 (0/33)	geometric shape	360	NA	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
11	Constable_2019_JEPHPP	Constable et al. (2019)	Exp1	Hungary	Hungarian	28 (12/16)	geometric shape	288	No License	
12	Constable_2019_JEPHPP	Constable et al. (2019)	Exp2	Hungary	Hungarian	28 (6/22)	geometric shape	288	No License	
13	Constable_2019_JEPHPP	Constable et al. (2019)	Exp3	Hungary	Hungarian	28 (8/14)	geometric shape	288	No License	
14	Constable_2019_JEPHPP	Constable et al. (2019)	Exp4	Hungary	English	40 (14/26)	geometric shape	640	No License	
15	Constable_2020_ActaPsych	Constable & Knoblich (2020)	Exp1	Hungary	Hungarian	48	geometric shape	240	No License	Lab Experiment
16	Constable_2020_ActaPsych	Constable & Knoblich (2020)	Exp1	Hungary	Hungarian	44	geometric shape	240	No License	Lab Experiment
17	Constable_2021_CogEmo	Constable et al. (2021)	Exp1	United Kingdom	English	28 (16/12)	face	384	No License	Lab Experiment
18	Constable_2021_CogEmo	Constable et al. (2021)	Exp1	United Kingdom	English	28 (9/19)	face	384	No License	Lab Experiment
19	Constable_2021_CogEmo	Constable et al. (2021)	Exp2	Hungary	English	28 (13/15)	grey scale squares	192	No License	Lab Experiment
20	Constable_2021_CogEmo	Constable et al. (2021)	Exp2	Hungary	English	28 (15/13)	grey scale squares	192	No License	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
21	Dalmaso_2024_ConsciousCog	Mario Dalmaso et al. (2024)	Exp1	Japan	Japanese	40 (19/21)	face	360	CC BY 4.0	Lab Experiment
22	Dalmaso_2024_ConsciousCog	Mario Dalmaso et al. (2024)	Exp2	Italy	Italian	40 (16/24)	face	360	CC BY 4.0	Lab Experiment
23	Feldborg_2021_IJERPH	M Feldborg et al. (2021)	Exp1	UK	English	53 (10/43)	geometri c shape and face	320	CC BY 4.0	Online Experiment
24	Feldborg_2021_IJERPH	M Feldborg et al. (2021)	Exp1	UK	English	49 (7/40)	geometri c shape and face	320	CC BY 4.0	Online Experiment
25	Golubickis_2021_ActaPsych	Golubickis & Macrae (2021)	Exp1	United Kingdo m	English	30 (9/21)	geometri c shape	360	NA	Online Experiment
26	Golubickis_2021_ActaPsych	Golubickis & Macrae (2021)	Exp2	United Kingdo m	English	25 (8/17)	geometri c shape	360	NA	Online Experiment
27	Haciahmet_2023_Psychophysiol	Haciahmet et al. (2023)	Exp1	German y	German	40 (8/32)	geometri c shape geometri c shape	240	CC BY 4.0	Lab Experiment
28	Hobbs_2023_PsychMed	Hobbs et al. (2023)	Exp1	United Kingdo m	English	144 (31/112)	c shape and face and	360	CC BY 4.0	Lab Experiment
29	Hu_2020_CollabraPsy	Hu et al. (2020)	Exp1	China	Chinese	44 (19/27)	currency geometri c shape	600	No License	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
30	Hu_2023_psyarxiv	Hu et al. (2023)	Exp3 a	China	Chinese	38 (23/15)	geometri c shape	720	CC-BY-4.0	
31	Hu_2023_psyarxiv	Hu et al. (2023)	Exp3 b	China	Chinese	61 (16/45)	geometri c shape	720	CC-BY-4.0	
32	Hu_2023_psyarxiv	Hu et al. (2023)	Exp4 a	China	Chinese	32 (17/15)	geometri c shape	360	CC-BY-4.0	
33	Hu_2023_psyarxiv	Hu et al. (2023)	Exp4 a	China	Chinese	32 (10/22)	geometri c shape	720	CC-BY-4.0	
34	Hu_2023_psyarxiv	Hu et al. (2023)	Exp6 b	China	Chinese	23 (15/8)	geometri c shape	2700-4320 (S1 1080 + S2 1620 + S3 0- 1620 per subject)	CC-BY-4.0	
35	Kirk_2025_BritJPsy	Neil W. Kirk et al. (2025)	Exp1	United Kingdom	English	35 (35/0)	voice	216	CC BY 4.0	Online Experiment
36	Kirk_2025_BritJPsy	Neil W. Kirk et al. (2025)	Exp2	United Kingdom	English	90 (90/0)	voice	216	CC BY 4.0	Online Experiment
37	Kolvoort_2020_HumBrainMap	IR Kolvoort et al. (2020)	Exp1	Canada	English	31 (17/14)	geometri c shape	100	NA	Lab Experiment
38	Lee_2023_Cognition	Lee et al. (2023)	Exp1	United Kingdom	English	47	geometri c shape	96	CC BY 4.0	Online Experiment
39	Lee_2023_Cognition	Lee et al. (2023)	Exp2	United Kingdom	English	51	geometri c shape	96	CC BY 4.0	Online Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
40	Lee_2026_BritJPsy	Lee et al. (2026)	Exp1a	United Kingdom	English	50 (27/23)	geometric shape	360	NA	Online Experiment
41	Lee_2026_BritJPsy	Lee et al. (2026)	Exp1b	United Kingdom	English	53 (19/34)	geometric shape	360	NA	Online Experiment
42	Lee_2026_BritJPsy	Lee et al. (2026)	Exp2	United Kingdom	English	50 (20/28)	geometric shape	360	NA	Online Experiment
43	Liang_2022_HumBrainMap	Liang et al. (2022)	Exp1	China	English	35 (13/22)	geometric shape	720	CC BY-NC-ND	Lab Experiment
44	Liang_2022_HumBrainMap	Liang et al. (2022)	Exp1	China	English	38 (22/16)	geometric shape	720	CC BY-NC-ND	Lab Experiment
45	Liang_2022_HumBrainMap	Liang et al. (2022)	Exp1	China	English	36 (13/23)	geometric shape	720	CC BY-NC-ND	Lab Experiment
46	Liu_2023_CogRes	Liu et al. (2023)	Exp1	UK	English	299 (132/161)	face	64	NA	Online Experiment
47	Martinez-Perez_2024_ConsciousCog	V Martínez-Pérez et al. (2024)	Exp2	Spain	Spanish	32 (5/27)	geometric shape	240	CC BY 4.0	Lab Experiment
48	Mcivor_2021_EJN	L McIvor et al. (2021)	Exp1	United Kingdom	English	20 (6/14)	geometric shape and face	960	CC BY 4.0	Lab Experiment
49	Mcivor_2021_EJN	L McIvor et al. (2021)	Exp1	United Kingdom	English	20 (2/18)	geometric shape and face	960	CC BY 4.0	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
50	Navon_2021_psyarxiv	Navon & Makovski (2021)	Exp1	Israel	Hebrew	13 (3/10)	geometric shape	360	CC0 1.0 Universal	Lab Experiment
51	Navon_2021_psyarxiv	Navon & Makovski (2021)	Exp2	Israel	Hebrew	27 (9/18)	geometric shape	360	CC0 1.0 Universal	Lab Experiment
52	Navon_2021_psyarxiv	Navon & Makovski (2021)	Exp3	Israel	Hebrew	28 (12/15)	geometric shape	360	CC0 1.0 Universal	Lab Experiment
53	Navon_2021_psyarxiv	Navon & Makovski (2021)	Exp4	Israel	Hebrew	27 (9/18)	geometric shape	360	CC0 1.0 Universal	Lab Experiment
54	Orellana-Corrales_2020_ExpPsych	G Orellana-Corrales et al. (2020)	Exp1	Germany	German	38 (9/29)	geometric shape	128	NA	Lab Experiment
55	Orellana-Corrales_2020_ExpPsych	G Orellana-Corrales et al. (2020)	Exp2	Germany	German	32 (2/3)	geometric shape	128	NA	Lab Experiment
56	Orellana-Corrales_2020_ExpPsych	G Orellana-Corrales et al. (2020)	Exp3	Germany	German	36 (7/29)	geometric shape	128	NA	Lab Experiment
57	Orellana-Corrales_2021_APP	G Orellana-Corrales et al. (2021)	Exp1	Germany	German	34 (9/25)	geometric shape	128	No License	Lab Experiment
58	Orellana-Corrales_2021_APP	G Orellana-Corrales et al. (2021)	Exp2	Germany	German	33 (11/23)	letter string	128	No License	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
59	Orellana-Corrales_2023_QJEP	G Orellana-Corrales et al. (2023)	Exp1	Germany	German	71 (46/25)	geometric shape	140	NA	Online Experiment
60	Orellana-Corrales_2023_QJEP	G Orellana-Corrales et al. (2023)	Exp1	Germany	German	65 (42/23)	geometric shape	140	NA	Online Experiment
61	Pan_2025_unpub	Pan (2025)	Exp1	China	Chinese	40	geometric shape		NA	
62	Perrykkad_2022_BMCPsych	Perrykkad et al. (2022)	Exp1	United States	English	334	geometric shape	360	CC BY 4.0	Online Experiment
63	Qi_2025_SciData	Qi (2025)	Exp1	China	Chinese	134 (57/77)	geometric shape	360	CC BY 4.0	Lab Experiment
64	Qian_2020_QJEP	Qian et al. (2020)	Exp1	China	Chinese	24 (9/15)	geometric shape	576	No License	Lab Experiment
65	Qian_2020_QJEP	Qian et al. (2020)	Exp2	China	Chinese	25 (7/18)	geometric shape	200	No License	Lab Experiment
66	Schaefer_2019_JCogPsych	Schaefer & Frings (2019)	Exp1	Germany	German	32	geometric shape	144	CC-BY-SA 4.0	Lab Experiment
67	Schaefer_2019_JCogPsych	Schaefer & Frings (2019)	Exp2	Germany	German	36	geometric shape	144	CC-BY-SA 4.0	Lab Experiment
68	Schaefer_2019_JCogPsych	Schaefer & Frings (2019)	Exp3	Germany	German	35	geometric shape	144	CC-BY-SA 4.0	Lab Experiment
69	Smith_2024_Cortex	Smith & Ferguson (2024)	Exp1	UK	English	48 (11/5)	face	360	CC BY-NC-ND	Lab Experiment
70	Sui_2014_APP	Sui et al. (2014)	Exp1	China	Chinese	24	geometric shape	360	NA	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
71	Sui_2014_APP	Sui et al. (2014)	Exp2	China	Chinese	18	geometri c shape	360	NA	Lab Experiment
72	Sui_2014_APP	Sui et al. (2014)	Exp3	China	Chinese	22	geometri c shape	360	NA	Lab Experiment
73	Sui_2014_APP	Sui et al. (2014)	Exp4	China	Chinese	20	geometri c shape	360	NA	Lab Experiment
74	Sui_2014_unpub	Sui (2014)	Exp1	China	Chinese	24	geometri c shape	720	NA	
75	Sui_2015_unpub	Sui (2015)	Exp1	China	Chinese	20	geometri c shape	960	NA	
76	Sui_2023_ConsciousCog	J Sui et al. (2023)	Exp1	UK	English	20 (12/8)	geometri c shape	960	CC BY- NC-ND	Lab Experiment
77	Sun_2026_DataExp	Sun et al (2026)	Exp1	China	Chinese	—	geometri c shape		No License	
78	Svensson_2022_PsychRes	Svensson et al. (2022)	Exp1	United Kingdo m	English	20 (6/14)	geometri c shape	200	NA	Lab Experiment
79	Svensson_2022_PsychRes	Svensson et al. (2022)	Exp2	United Kingdo m	English	24 (8/16)	geometri c shape	400	NA	Lab Experiment
80	Svensson_2022_PsychRes	Svensson et al. (2022)	Exp3	United Kingdo m	English	25 (7/18)	geometri c shape	400	NA	Lab Experiment
81	Svensson_2023_QJEP	Svensson et al. (2023)	Exp1	UK	English	65 (20/45)	geometri c shape	120	No License	Online Experiment
82	Vicovaro_2022_JEPHPP	Vicovaro et al. (2022)	Exp1	Italy	Italian	30 (14/16)	random dots	480	CC BY 4.0	Mixed (Lab + Online)

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
83	Vicovaro_2022_JEPHPP	Vicovaro et al. (2022)	Exp2	Italy	Italian	52 (11/36)	random dots	240	CC BY 4.0	Mixed (Lab + Online)
84	Vicovaro_2022_JEPHPP	Vicovaro et al. (2022)	Exp2	Italy	Italian	52 (5/42)	random dots	240	CC BY 4.0	Mixed (Lab + Online)
85	Vicovaro_2024_PeerJ	Vicovaro et al. (2024)	Exp1	Italy	English	64 (26/38)	geometri c shape	240	NA	Lab Experiment
86	Vicovaro_2024_PeerJ	Vicovaro et al. (2024)	Exp2	Italy	English	64 (16/48)	geometri c shape	240	NA	Lab Experiment
87	Wang_2016_JEPHPP	Wang et al. (2016)	Exp1	China	English	21 (3/18)	geometri c shape	648	CC BY 3.0	
88	Wang_2016_JEPHPP	Wang et al. (2016)	Exp2	China	English	25 (10/15)	geometri c shape	648	CC BY 3.0	
89	Wozniak_2018_PLOS	Wozniak et al. (2018)	Exp1	Hungary	English	18 (9/9)	face	672	CC BY 4.0	Lab Experiment
90	Wozniak_2018_PLOS	Wozniak et al. (2018)	Exp2	Hungary	English	18 (9/9)	face	672	CC BY 4.0	Lab Experiment
91	Wozniak_2020_PLOS	Wozniak et al. (2020)	Exp1	Australi a	English	24 (12/12)	face	270	NA	Lab Experiment
92	Wozniak_2020_PLOS	Wozniak et al. (2020)	Exp2	Australi a	English	24 (12/12)	face	270	NA	Lab Experiment
93	Wozniak_2020_PLOS	Wozniak et al. (2020)	Exp3	Australi a	English	24 (12/12)	face	270	NA	Lab Experiment
94	Wozniak_2022_PsychRes	Wozniak et al. (2022)	Exp1	Hungary	English	18 (7/11)	geometri c shape	720	CC BY 4.0	Lab Experiment
95	Wozniak_2022_PsychRes	Wozniak et al. (2022)	Exp2	Hungary	English	18 (6/12)	geometri c shape	720	CC BY 4.0	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
96	Wozniak_2022_PsychRes	Wozniak et al. (2022)	Exp3	Hungary	English	18 (0/18)	geometric shape	720	CC BY 4.0	Lab Experiment
97	Xu_2022_CurrPsych	Xu et al. (2022)	Exp1	China	Chinese	55 (27/28)	geometric shape	360	On request	Lab Experiment
98	Xu_2022_CurrPsych	Xu et al. (2022)	Exp1	China	Chinese	50 (24/26)	geometric shape	360	On request	Lab Experiment
99	Zhang_2023_NeuroImage	Yongfa Zhang et al. (2023)	Exp1	China	Chinese	346 (172/174)	geometric shape	96	CC BY-NC-ND	Lab Experiment
100	Zhang_2024_PsychJ	Zhang et al. (2024)	Exp1	China	English	43	geometric shape	216	NA	Lab Experiment
101	Zhang_2024_PsychJ	Zhang et al. (2024)	Exp2	China	English	36	geometric shape	120	NA	Lab Experiment
102	Zhang_2026_JNeurosci	Zhang (2026)	Exp1	China	English	59 (27/32)	geometric shape	450 (5 blocks x 90; incomplete blocks in 17 subjects)	NA	Lab Experiment
103	Zhang_2026_JNeurosci	Zhang (2026)	Exp1	China	English	32 (14/18)	geometric shape	450 (5 blocks x 90; incomplete blocks in 17 subjects)	NA	Lab Experiment
104	Zhao_2026_PsychonBullRev	Zhao & Wang (2026)	Exp1	China		45 (12/33)	geometric shape	360	NA	Lab Experiment
105	Zhao_2026_PsychonBullRev	Zhao & Wang (2026)	Exp2	China		47 (18/29)	geometric shape	360	NA	Lab Experiment
106	Zhao_2026_PsychonBullRev	Zhao & Wang (2026)	Exp3	China		22 (12/10)	geometric shape	360	NA	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
107	Zhao_2026_PsychonBullRev	Zhao & Wang (2026)	Exp3	China		22 (13/9)	geometri c shape	360	NA	Lab Experiment

Note. N = number of participants retained in the minimally preprocessed data of each experiment-by-participant-group entry; Trials = number of test trials per participant as documented in the master index; Exp_Implement = laboratory or online administration as documented in the experiment-level JSON metadata. The diversity of countries, languages, stimulus modalities, and experimental designs ensures that future re-analyses can address a broad range of scientific questions—from the robustness and generalisability of the SPE across cultures and stimulus formats, to the identification of experimental design parameters that moderate the magnitude of the effect.

As we mentioned in the section Folder Structure, data files include both raw trial-level data files shared from each team (e.g., "Amodeo_2024_CABN_Exp1_raw.csv") and minimally preprocessed data files (e.g., "Amodeo_2024_CABN_Exp1_Clean.csv"), both are stored in experiment-specific subfolders (see Section XXX). The corresponding codebook (e.g., "Codebook_Amodeo_2024_CABN_Exp1_Clean.xlsx") describes each column (see Figure 3). Basic information such as participant index (Subject), matching condition (Matching; matching or nonmatching), shape stimulus (Shape), label stimulus (Label), reaction time in milliseconds (RT_ms) and seconds (RT_sec), and response accuracy (ACC) are included in all experiments. The identity columns for both shape and label in data files were standardized into six categories: Self, Close, Acquaintance, Celebrity, Stranger, and NonPerson (Sun et al., 2023). For clarity, we kept the original identity names ("Origin_Identity"), the English-translated identity names ("English_Identity"), and the standardised identity categories ("Standardised_Identity"). The minimally preprocessed data files may also include additional fields that describe the experimental design, such as block number, trial number within block, experimental phase or session, specific stimulus features (e.g., face gender, voice accent, emotional valence), and participant response keys. We expect these minimally preprocessed data files will be valuable for researchers interested in re-using this database.

Variable_name	Variable_description	Variable_value	Variable_category
Subject	Participant number	Number	Numerical
Block	Block number in the experiment; each block may contain practice or experimental trials	Number	Numerical
Trial	Trial number within the block	Number	Numerical
Shape	Visual shape stimulus presented in the trial	T. bmp;S. bmp;C. bmp	Categorical
Label	Text label presented for matching with the shape	Vreemde;Ikzelf;Bekende	Categorical
Matching	Type of matching task for the trial; indicates whether the shape and label match or not	Matching;Nonmatching	Categorical
Label_Origin_Identity	Original identity associated with the label	Vreemde;Ikzelf;Bekende	Categorical
Label_English_Identity	English translation of the label's identity	Stranger;Self;Friend	Categorical
Label_Standardized_Identity	Standardized identity for the label used across experiments	Stranger;Self;Friend	Categorical
Shape_Origin_Identity	Original identity associated with the shape stimulus	Stranger;Self;Friend	Categorical
Shape_English_Identity	English translation of the shape's identity	Stranger;Self;Friend	Categorical
Shape_Standardized_Identity	Standardized identity for the shape used across experiments	Stranger;Self;Close	Categorical
Response	Participant's response in the trial	n;m	Categorical
RT_ms	Reaction time for the response, measured in milliseconds	Number	Numerical
RT_sec	Reaction time for the response, measured in seconds	Number	Numerical
ACC	Accuracy of the participant's response	0;1	Categorical

Figure 3. Standardized codebook of the experimental dataset variables.

Example uses of the SPE Database

The Self-Prioritization Database provides a valuable resource for both methodological and theoretical investigations of the self-prioritization effect. For example, the standardized schema allows researcher to develop and test cognitive models of self-prioritization effects; investigate how experimental design parameters shape these effects; clarify the cross-cultural universality and cultural specificity of self-prioritization; and evaluate the robustness and generalizability of findings across laboratory-based and online settings. To demonstrate the unique value of large-scale standardized datasets, we present three examples addressing questions that cannot be adequately examined using data from individual studies alone.

The Magnitude of Self-Prioritization Effect Varies Across Social Distance

The self-prioritization effect (SPE) is typically quantified as the performance difference between self-referential and other-referential conditions. However, different categories of others have been used as baseline conditions when estimating SPE magnitude. Existing studies have yielded inconsistent estimates of the self-prioritization effect (SPE) magnitude. Nevertheless, a systematic examination of how baseline selection affects SPE magnitude remains lacking. Leveraging the large-scale sample size of the SPE database, we compared the effect size of the SPE across five standardized baselines (Sun et al., 2023): Non-person, Stranger, Celebrity, Acquaintance, and Close Other. Standardized effect sizes (Cohen's d) were computed at the participant level by contrasting self-related trials against each baseline condition separately, using a pooled standard deviation. Reaction-time based effect sizes were computed on correct trials with response times above 0 ms and no longer than 10,000 ms; accuracy-based effect sizes were computed on trials with a valid response (0 = error, 1 =

correct). Because participant identifiers are unique only within a dataset, participants were nested within datasets (that is, the random effect was defined on the dataset-by-participant key). A linear mixed-effects model without intercept was then fitted to the participant-level Cohen's d values, with this nested participant term as a random intercept and data source as an additional variance component. From this model, 10,000 multivariate normal samples were drawn from the estimated fixed-effects distribution to obtain the posterior-like distributions of each identity's SPE magnitude. For consistency, we use positive Cohen's d for SPE (i.e., faster reaction times or higher accuracy for self relative to other identities).

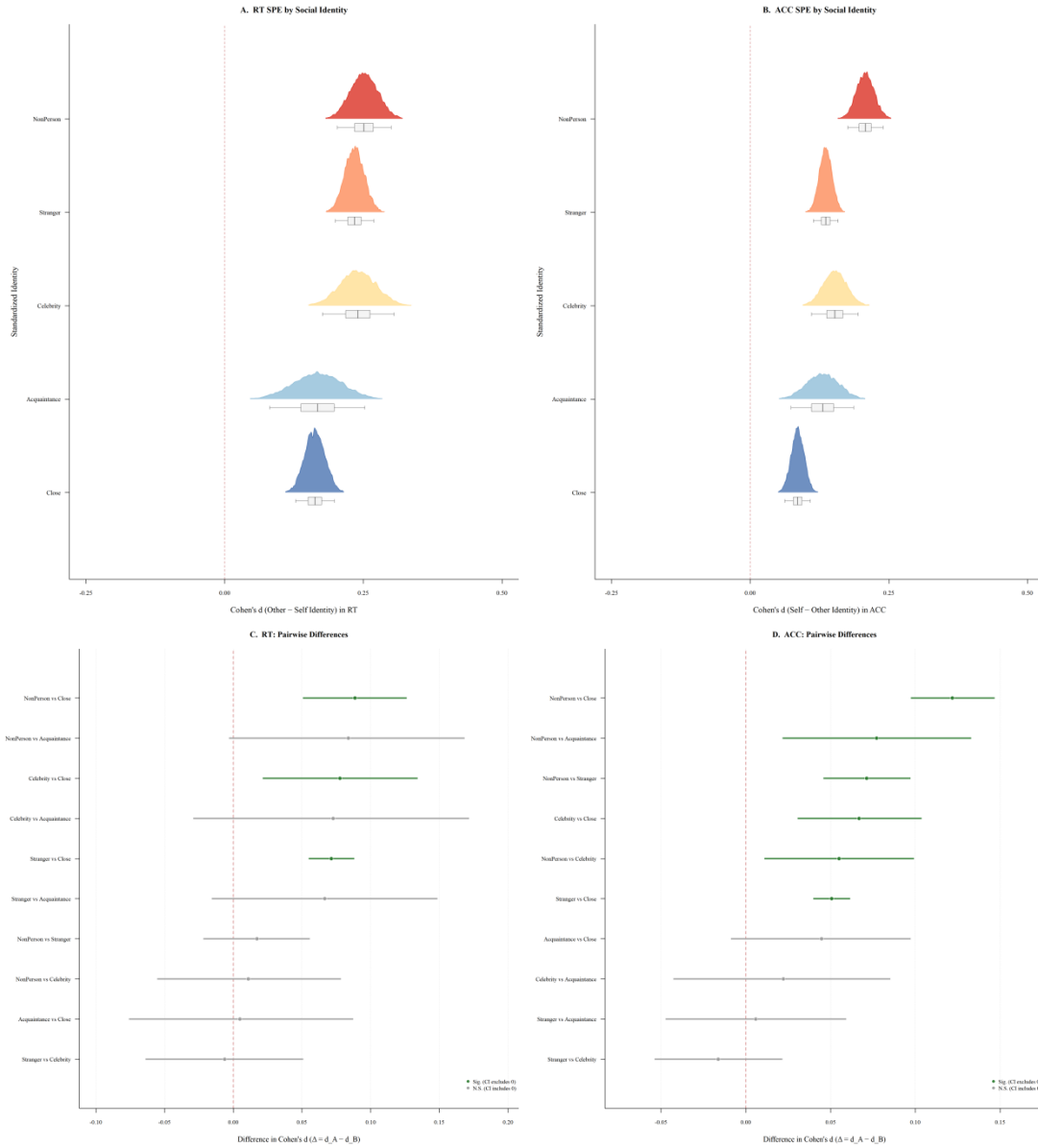


Figure 4. Self-prioritization effects along the social distance continuum and pairwise comparisons across social identities.

For reaction time (RT), the SPE was positive for all five baseline conditions: every bootstrap distribution of the participant-level Cohen's d lay entirely above zero (all 95% confidence intervals excluded zero; see Figure 4A). The magnitude of the SPE, however, differed only between some baselines (see Figure 4). To test whether the magnitude differed statistically between SPEs calculated with different baselines, we computed these effect sizes

pairwise ($\Delta = dA - dB$) using the same model-based sampling approach, and inferred a difference when the bootstrap 95% CI excluded zero (see Figure 4C and 4D).

For RT, 3 of the 10 pairwise comparisons were statistically significant, and all three involved Close others as the baseline. The Non-person-based SPE was larger than the Close-other-based SPE ($\Delta = 0.089$, 95% CI [0.051, 0.126], $p < .001$); the Celebrity-based SPE was larger than the Close-other-based SPE ($\Delta = 0.078$, 95% CI [0.022, 0.134], $p = .007$); and the Stranger-based SPE was larger than the Close-other-based SPE ($\Delta = 0.071$, 95% CI [0.055, 0.088], $p < .001$). The remaining seven comparisons were not statistically significant, including all comparisons among Non-person, Stranger, and Celebrity (all $|\Delta| \leq 0.018$, all $ps \geq .372$) and the Acquaintance-versus-Close-other comparison ($\Delta = 0.005$, 95% CI [-0.076, 0.087], $p = .910$).

Six of the 10 pairwise comparisons reached statistical significance for accuracy (ACC). The Non-person-based SPE was larger than that based on Close others ($\Delta = 0.122$, 95% CI [0.098, 0.146], $p < .001$), Acquaintances ($\Delta = 0.077$, 95% CI [0.022, 0.133], $p = .006$), Strangers ($\Delta = 0.071$, 95% CI [0.046, 0.097], $p < .001$), and Celebrities ($\Delta = 0.055$, 95% CI [0.011, 0.099], $p = .014$). The Celebrity-based SPE was larger than the Close-other-based SPE ($\Delta = 0.067$, 95% CI [0.031, 0.103], $p < .001$), and the Stranger-based SPE was larger than the Close-other-based SPE ($\Delta = 0.051$, 95% CI [0.040, 0.061], $p < .001$). The remaining four comparisons were not significant: Stranger versus Celebrity ($\Delta = -0.016$, 95% CI [-0.054, 0.021], $p = .386$), Stranger versus Acquaintance ($\Delta = 0.006$, 95% CI [-0.047, 0.059], $p = .827$), Celebrity versus Acquaintance ($\Delta = 0.022$, 95% CI [-0.042, 0.085], $p = .494$), and Acquaintance versus Close other ($\Delta = 0.045$, 95% CI [-0.008, 0.097], $p = .096$).

Taken together, the estimated magnitude of the SPE was reliably positive against every baseline, but the baselines did not order themselves along a clear social-distance gradient. Across both RT and ACC, the only consistent contrast was between the baselines with the largest and the smallest point estimates: the Non-person-based (and, for RT, also the Stranger- and Celebrity-based) SPE exceeded the Close-other-based SPE, whereas the comparisons among Non-person, Stranger, and Celebrity were not reliable in either measure. Two caveats are important. First, the baselines differed markedly in coverage: Non-person and Celebrity were represented by only 8 and 3 datasets respectively, whereas Stranger and Close others were represented by 70 and 50 datasets, so the absence of reliable differences among the more sparsely covered baselines should be interpreted with caution. Second, this pattern qualifies rather than confirms the proposal that the SPE follows a monotonic gradient of closeness to the self (Aron et al., 1992; Markus & Kitayama, 1991).

Presence of Self-Prioritization Effect Under Nonmatching Conditions

Previous studies have reported inconsistent findings regarding the presence of self-prioritization effects (SPEs) under nonmatching conditions. Although several studies have observed reliable nonmatching SPEs (Kirk & Cunningham, 2025), others have failed to detect such effects (Desebrock & Spence, 2021; Golubickis & Macrae, 2021). One potential explanation for these inconsistencies is that nonmatching SPEs are substantially smaller than matching SPEs and therefore require larger samples to be detected reliably. An alternative is that the SPEs under nonmatching conditions were contaminated by the involvement of self-reference in the baseline condition that used for calculating SPE.

Leveraging the large sample size of the current database (89 experiment-level datasets, 1,068,081 nonmatching trials, 4,520 participants), we estimated the SPE using a bootstrap resampling approach (500 iterations per sample size, with replacement) that allows evaluation of effect size stability across increasing sample sizes. Recall that in the self-matching task, each trial presents a shape–label pair. Thus, for matching trials, the shape and label are congruent (e.g., a shape previously associated with "Self" paired with the word "Self"), making the distinction between self-related and other-related trials straightforward. For nonmatching conditions (also referred to as incongruent or mismatching trials), however, the shape and label convey conflicting identity information (e.g., a shape for "Self" paired with the word "Stranger"), creating difficulty in defining what constitutes a "Self" trial. Reaction-time based Cohen's *d* values were computed on correct trials with response times above 0 ms and no longer than 10,000 ms, accuracy-based Cohen's *d* values on trials with a valid response (0 = error, 1 = correct), and participants were again identified by the dataset-by-participant key.

To address this ambiguity, we adopted two alternative operationalizations of the self-prioritization effect (SPE) under nonmatching conditions: shape-based SPE or label-based SPE. **The shape-based SPE is the difference** between trials where the shape is "Self" and the label is non-Self and trials where the shape is non-self condition(s) and the label does not match the shape. **The label-based SPE is difference** between trials where the label is "Self" and the shape is non-self and trials where the label is non-self condition(s) and the shape does not match the label.

Critically, neither the shape-based nor the label-based operationalization guarantees a clean self versus non-self contrast, because nonmatching trials classified as "non-self" may still

contain self-referential information. For example, in an experiment includes only two identities, Self and Stranger, the nonmatching trials are either "Self(shape)-Stranger(label)" or "Stranger(shape)-Self(label)". Under both conditions, part of the information is self-referential, making it impossible to construct a pure contrast between self- versus non-self-referential conditions. A pure self versus non-self contrast becomes feasible only in experiments involving three or more identities. For example, under the shape-based operationalization, self-related trials can be defined as "Self(shape)-Other1(label)/Other2(label)" whereas non-self trials can be defined as "Other1(shape)-Other2(label)". In this case, the comparison trials contain no self-referential information, thereby providing a rigorous estimate of nonmatching SPEs.

To fully exploit the available datasets while preserving interpretability, we implemented both conservative and liberal approach for estimating the SPE for nonmatching trials. The conservative approach only include pure self versus non-self contrasts: only participants from experiments involving at least three distinct identities were eligible for inclusion. At the trial level, additional filtering procedures were applied. For shape-based SPE, nonmatching trials in which the shape corresponded to a non-self identity but the label referred to the Self were excluded. Conversely, for label-based SPE, trials in which the label corresponded to a non-self identity but the shape was associated with the Self were removed. Using these criteria, 38 datasets contributed to the conservative shape-based analysis (1,977 eligible participants) and 27 datasets to the conservative label-based analysis (1,565 eligible participants). The final datasets comprised 371,894 trials for the shape-based SPE and 276,010 trials for the label-based SPE. The liberal approach retained 85 datasets (5,149 participants, 724,323 trials) for the shape-based SPE and 69 datasets (4,487 participants, 628,024 trials) for the label-based SPE.

In contrast, the liberal approach maximized dataset coverage by retaining all eligible experiments and nonmatching trials regardless of the presence of residual self-referential information in the comparison condition. To ensure a fair comparison between shape-based and label-based SPE for nonmatching trials, we aligned sample-size by using the minimum N of the shape- and label-based SPE for the conservative approach (see Table 2 for the bootstrap sample sizes and the results of the two approaches).

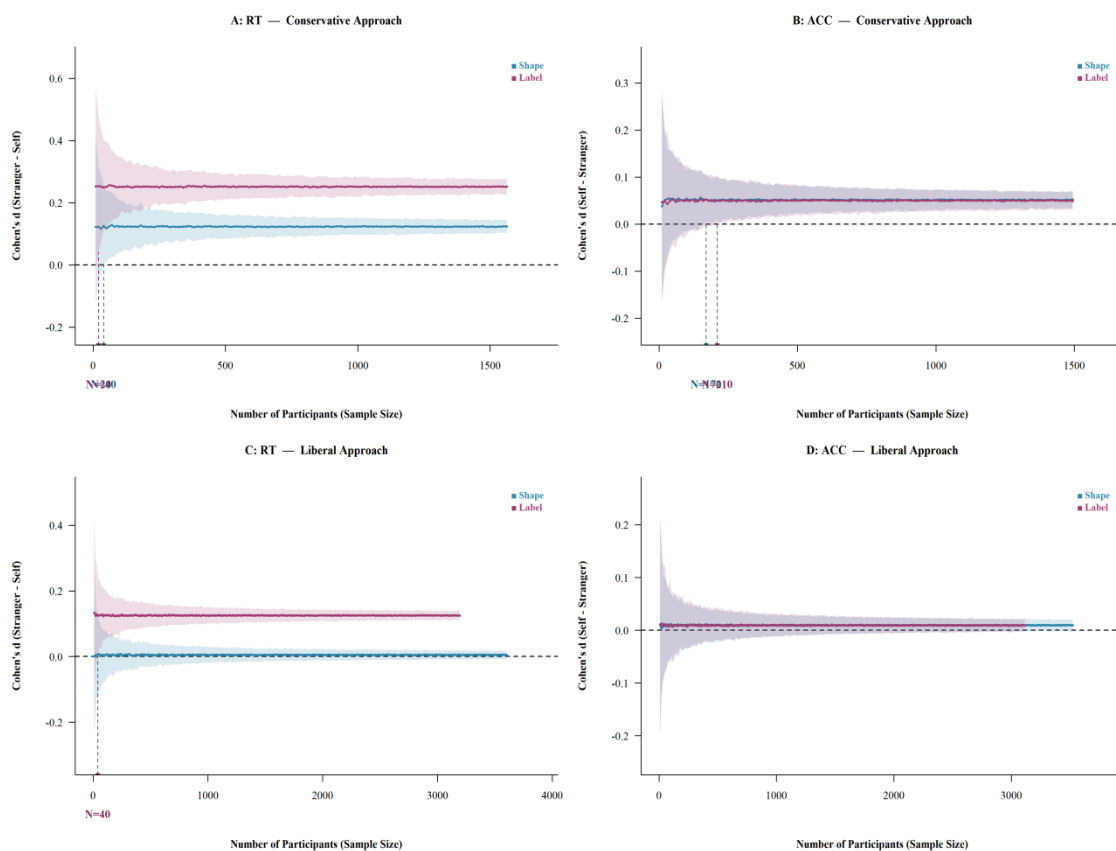


Figure 5. Bootstrap estimation of the self-prioritization effect under nonmatching conditions. A and B, the conservative approach; C and D, the liberal approach.

Table 2

Bootstrap Estimates of the Self-Prioritization Effect Under Nonmatching Conditions

	Measure	Dimension	N _{Max}	Cohen's d	95% CI	N _{min}
Liberal Approach	RT	Shape	3,604	0.005	[-0.008, 0.017]	not reached
	RT	Label	3,192	0.125	[0.112, 0.138]	40

Conservative Approach	ACC	Shape	3,524	0.009	[−0.002, 0.020]	not reached
	ACC	Label	3,114	0.009	[−0.003, 0.021]	not reached
	RT	Shape	1,563	0.123	[0.103, 0.144]	40
	RT	Label	1,563	0.252	[0.227, 0.275]	20
	ACC	Shape	1,494	0.052	[0.035, 0.070]	170
	ACC	Label	1,494	0.049	[0.030, 0.069]	210

Note. For RT, $d = (\text{Mean_Stranger} - \text{Mean_Self}) / \text{Pooled_SD}$, such that positive values indicate faster responses to self trials (SPE). For ACC, $d = (\text{Mean_Self} - \text{Mean_Stranger}) / \text{Pooled_SD}$, such that positive values indicate higher accuracy for self trials (SPE). $N_{\min}$ is the smallest bootstrap sample size at which the 95% CI no longer includes zero, as marked by vertical dashed lines in Figure 5; "not reached" indicates that the CI included zero across the full range of sample sizes. For the conservative approach, $N_{\max}$ is the sample size aligned across the shape-based and label-based analyses.

Using the conservative approach, a robust SPE was observed for both reaction time (RT) and accuracy (ACC). For RT, the label-based operationalization yielded a moderate SPE ($d = 0.252$, 95% CI [0.227, 0.275], $N = 1,563$), indicating faster responses for nonmatching trials in which the label corresponded to the Self relative to those in which the label referred to other identities. The corresponding 95% confidence interval first excluded zero at merely $N = 20$, indicating that the label-based RT SPE can be detected even in relatively small samples (see Figure 5A). The shape-based operationalization also revealed a significant RT SPE, although the effect was substantially smaller ($d = 0.123$, 95% CI [0.103, 0.144], $N = 1,563$); in this case the 95% confidence interval first excluded zero at $N = 40$ (see Figure 5A). For accuracy, both operationalizations produced relatively small but statistically significant effects. The shape-based SPE was estimated at $d = 0.052$, 95% CI [0.035, 0.070], $N = 1,494$, with the confidence interval first excluding zero at $N = 170$ (see Figure 5B). The label-based SPE was comparable,

$d = 0.049$, 95% CI [0.030, 0.069], $N = 1,494$, with the confidence interval first excluding zero at $N = 210$ (see Figure 5B). Taken together, a SPE was present for both RT and ACC for nonmatching trials when the conservative approach was used. Descriptively, the label-based SPE was about twice as large as the shape-based SPE for RT (0.252 vs 0.123), whereas the two operationalizations were nearly indistinguishable for ACC (0.049 vs 0.052).

When applying the liberal approach, the label-based SPE was again evident for reaction time (RT), although the effect size was smaller than that observed under the conservative approach, $d = 0.125$, 95% CI [0.112, 0.138], $N = 3,192$. The corresponding 95% confidence interval first excluded zero at $N = 40$ (see Figure 5C), indicating that the label-based RT effect remained detectable with relatively modest sample sizes. In contrast, the shape-based SPE was markedly attenuated under the liberal approach: the point estimate was close to zero ($d = 0.005$, 95% CI [-0.008, 0.017], $N = 3,604$) and the confidence interval never excluded zero across the full range of sample sizes (see Figure 5C), suggesting that the effect was too small to be detected reliably in typical experimental samples. For accuracy, neither operationalization was reliable under the liberal approach: shape-based, $d = 0.009$, 95% CI [-0.002, 0.020], $N = 3,524$; label-based, $d = 0.009$, 95% CI [-0.003, 0.021], $N = 3,114$; neither confidence interval excluded zero at any sample size (see Figure 5D).

Collectively, the presence of a SPE on nonmatching trials depended both on the operationalization of self-relevance and on how strictly the comparison conditions were purified. When only pure self versus non-self contrasts were admitted (conservative approach), the SPE was reliably present for both RT and ACC; when all nonmatching trials were retained (liberal approach), only the label-based RT effect survived. The conservative-liberal

discrepancy indicates that part of the apparent nonmatching SPE in the liberal analysis is attributable to residual self-referential information in the comparison trials, exactly as anticipated above. The RT effects were consistently larger than the ACC effects, a pattern that is consistent with previous findings in the SPE literature (Liu et al., 2025). These findings underscores the importance of both adequate statistical power and transparent operational definitions when investigating the SPE under nonmatching conditions.

The Association Between Presenting Duration / Trial Number and magnitude of SPE

The third example illustrates a unique advantage of the current database: the standardized coding of methodological variations in task implementation across studies. Specifically, we examined two implementation parameters that varied across experiments: stimulus presentation duration and trial number. Stimulus duration was chosen because shorter presentation times may create time pressure for participants and alter their processing strategies (Lemaire & Brun, 2016; Ratcliff et al., 2016). Despite its potential theoretical relevance, variation in stimulus duration has received little attention in the SPE literature and is often treated as a minor procedural detail. Trial number was examined because the amount of behavioral data collected from each participant may influence the reliability and stability of SPE estimates, consistent with broader concerns regarding measurement precision in cognitive tasks (Liu et al., 2025). Both implementation parameters were treated as continuous variables. For each analysis, we estimated the association between the implementation parameter and SPE magnitude (Cohen's d_z , computed against the Stranger baseline) using bootstrap Spearman correlations with 10,000 resamples. We report the observed Spearman ρ , bootstrap 95% confidence intervals (2.5th – 97.5th percentiles), and bootstrap p values (two-tailed; statistical

significance assessed at $\alpha = .05$). Two datasets were excluded before analysis (Pan_2025_unpub and Wang_2016_JEPHPP), and datasets whose Stranger-based Cohen's d_z deviated from the mean by more than three standard deviations were treated as outliers and removed.

The association between stimulus presentation duration and SPE magnitude was negative. Treating duration as a continuous variable (range: 100 – 3,000 ms, $n = 54$ datasets), bootstrap Spearman correlations revealed a significant negative relationship for both RT ($\rho = -.311$, bootstrap 95% CI $[-.539, -.052]$, $p_{\text{boot}} = .018$) and ACC ($\rho = -.335$, bootstrap 95% CI $[-.548, -.095]$, $p_{\text{boot}} = .010$; Figure 6A and 6B). Both bootstrap confidence intervals lay entirely below zero, providing evidence for a reliable negative association between stimulus duration and the magnitude of self-prioritization. Shorter presentation times were associated with larger SPE effect sizes, whereas longer presentation times tended to yield smaller effects.

In contrast, the maximum number of trials in a dataset was not reliably associated with SPE magnitude (Figure 6C and 6D). For reaction time (RT), the bootstrap Spearman correlation was close to zero ($\rho = .045$, bootstrap 95% CI $[-.222, .302]$, $p_{\text{boot}} = .748$); for accuracy (ACC), the point estimate was numerically positive but the confidence interval again included zero ($\rho = .174$, bootstrap 95% CI $[-.098, .433]$, $p_{\text{boot}} = .205$). To check whether these associations were driven by a small number of datasets with extreme trial counts, we conducted a sensitivity analysis excluding the three datasets with the largest trial numbers; the picture was unchanged (RT: $\rho = .077$, $p = .560$; ACC: $\rho = .195$, $p = .132$). The number of trials a dataset contains was therefore not a reliable moderator of the SPE in the present database.

Together, these exploratory findings indicate that the magnitude of the SPE varies with

how the task is implemented, but that not every candidate implementation parameter is a reliable moderator. Shorter stimulus presentation durations were associated with larger SPE magnitudes, and this held for both RT and ACC, whereas the maximum number of trials was unrelated to SPE magnitude. These findings should be interpreted with caution. The relationship between SPE magnitude and stimulus presentation duration may not be strictly linear and may depend on important boundary conditions. For example, the SPE may change when the stimulus presentation duration becomes subliminal. Likewise, the absence of a trial-number effect does not exclude non-linear or plateau-like relationships, and the available range of trial numbers is confounded with other study-level design choices. Future studies are needed to experimentally manipulate these implementation parameters to establish their causal contributions to SPE magnitude. Nevertheless, our findings highlight the importance of considering methodological details when studying self-prioritization. More broadly, they demonstrate the value of systematically coding task implementation characteristics in large-scale databases: variables that are often treated as minor procedural details may meaningfully contribute to variability in observed SPE estimates across studies, and standardized metadata are what make such questions answerable at all.

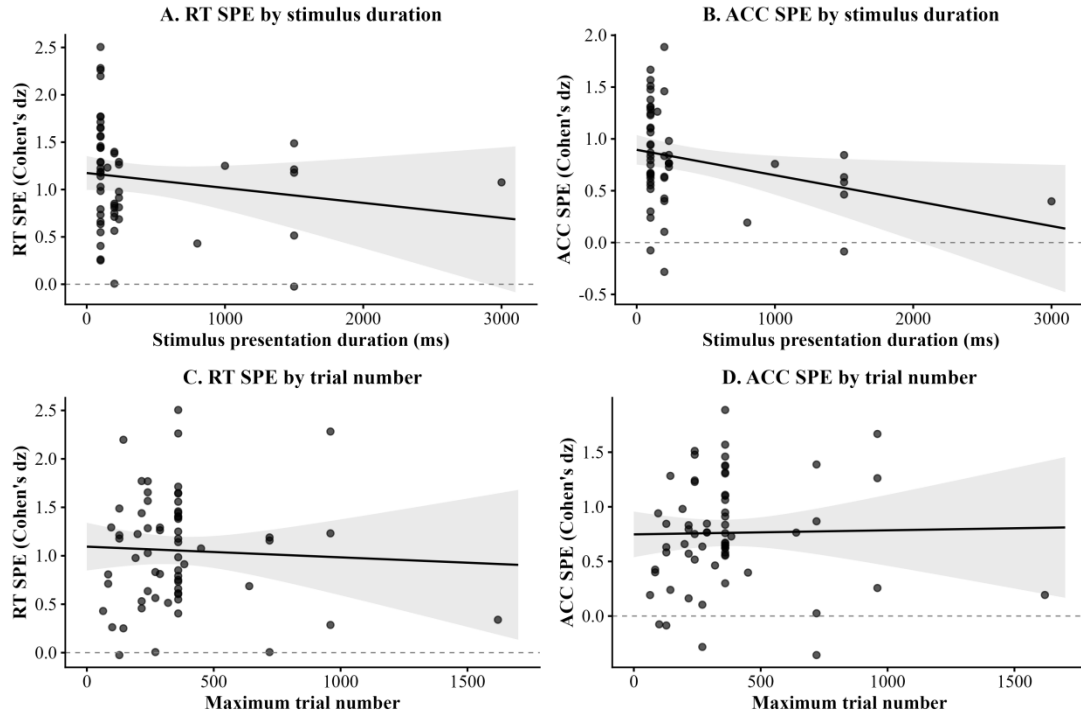


Figure 6. Exploratory moderators of the self-prioritization effect under matching conditions.

Discussion

The present study introduces both standardized data organization framework for self-matching task and the first manually curated, large-scale standardized database for the self-prioritization effect (SPE). Guided by the FAIR principles of data management (Wilkinson et al., 2016), we developed a three-level folder structure consisting of root-, study-, and experiment-level folders, each included standardized, machine-readable JSON metadata files. This folder structure strikes a balance between human interpretability and machine readability, thereby facilitating data sharing, reuse, and large-scale secondary analyses. Moreover, we developed a standardized, five component framework for annotating the metadata of task implementation details of self-matching task, which will facilitate cumulative, reproducible, and data-intensive research on self-prioritization. By applying the folder structure and

standardized framework for annotating task details to available data, we curated a large database with unified meta data, which allows researchers to address theoretical and methodological questions that would be difficult or impossible to examine using individual studies alone.

The SPE database provides a valuable resource for understanding of SPE. Through three illustrative examples, we demonstrated how standardized, large-scale datasets can address theoretical and methodological questions that can not be solved by single small sample study. We found that the SPE was reliably positive against every baseline condition but that the baselines did not order themselves along a simple social-distance gradient; that the presence of a SPE under nonmatching conditions depended on how self-relevance was operationalized; and that the magnitude of the SPE was related to stimulus presentation duration but not to the number of trials. Together, these findings exemplify the value of such a well-maintained large-scale database for the field.

It worth noting that the SPE database provides the first standardized framework for annotation of data generated from self-matching task. We decomposes the implementation of the self-matching task into five core components: Physical Environment, Experimental Design, Block Structure, Trial Structure, and Stimulus Properties. This decomposition follows explicit operational criteria grounded in classical information-processing theory (Gold & Shadlen, 2007) and formal decision-making models (Ratcliff et al., 2016), thereby providing clear boundaries for assigning variables to each component and avoiding arbitrary categorization. By including comprehensive details in metadata, our database substantially enriches the methodological information available to researchers in ways that are not typically provided in

published articles, as demonstrated by our third example.

Importantly, this standardized framework can extend beyond the self-matching task. Many laboratory-based cognitive tasks share similar structural components, our framework can serve as a template for standardizing metadata regarding the implementation details in other tasks. Reporting these details in machine-readable format will increase the transparency of empirical studies and provide more nuanced data beyond the current best practices recommended by Journal Article Reporting Standards for quantitative research in psychology (Appelbaum et al., 2018). These rich information will be valuable for both peer-review process, post-publication review, and meta-research. The current cognitive tasks may also vulnerable to the jingle-jangle fallacy (e.g., Sun, Wang, Wen, & Chuan-Peng, 2023), reporting these details will help to understand how the details of implementing task will change the cognitive construct under the investigation.

Limitations of the current database should be acknowledged. First, the curation of SPE database relied heavily on manual extraction, limiting the scalability and efficiency of the standardized framework for task implementation. Future efforts may learn from existing standards such as the Brain Imaging Data Structure (BIDS, Poldrack et al., 2024) or Hierarchical Event Descriptors (HED, Robbins et al., 2022) to improve the level of. Second, the current version of SPE database focuses exclusively on behavioral measures. Future versions should aim to include multimodal datasets, such as EEG (e.g., Sui et al., 2023), fMRI (e.g., Sui et al., 2013), using established standards for neuroimaging data organization. Third, the current SPE database only included data that are permitted by authors or the data license, potentially introducing bias into the database, such as the participants' representativeness (Ghai

et al., 2024; Ghai et al., 2025; Liu et al., 2025). We made all code and script of the present database open available to facilitate future larger-scale collaborative projects that will expand the SPE database.

To conclude, we believe the SPE database will serve as a value resource for community-driven initiatives including large-scale collaborative projects and crowdsourced replication efforts. By combining standardized data structures with comprehensive methodological metadata, the database provides an infrastructure capable of supporting coordinated investigations of self-prioritization across diverse experimental settings and populations. More broadly, this work illustrates how standardized, FAIR-compliant databases can transform fragmented collections of open datasets into cumulative scientific resources that accelerate theory development, improve reproducibility, and promote collaborative discovery in cognitive psychology.

Reference

- Appelbaum, M., Cooper, H., Kline, R. B., Mayo-Wilson, E., Nezu, A. M., & Rao, S. M. (2018). Journal article reporting standards for quantitative research in psychology: The APA publications and communications board task force report. *American Psychologist*, 73(1), 3–25. <https://doi.org/10.1037/amp0000191>
- Aron, A., Aron, E. N., & Smollan, D. (1992). Inclusion of Other in the Self Scale and the structure of interpersonal closeness. *Journal of Personality and Social Psychology*, 63(4), 596–612. <https://doi.org/10.1037/0022-3514.63.4.596>
- Bogacz, R., Brown, E., Moehlis, J., Holmes, P., & Cohen, J. D. (2006). The physics of optimal decision making: A formal analysis of models of performance in two-alternative forced-choice tasks. *Psychological Review*, 113(4), 700–765. <https://doi.org/10.1037/0033-295X.113.4.700>
- Bridges, D., Pitiot, A., MacAskill, M. R., & Peirce, J. W. (2020). The timing mega-study: Comparing a range of experiment generators, both lab-based and online. *PeerJ*, 8, e9414. <https://doi.org/10.7717/peerj.9414>
- Cheung, M. W.-L., & Jak, S. (2016). Analyzing big data in psychology: A split/analyze/meta-analyze approach. *Frontiers in Psychology*, 7, 1–13. <https://doi.org/10.3389/fpsyg.2016.00738>
- Cicchini, G. M., Mikellidou, K., & Burr, D. C. (2024). Serial Dependence in Perception. *Annual Review of Psychology*, 75(1), 129–154. <https://doi.org/10.1146/annurev-psych-021523-104939>
- Constable, M. D., Rajsic, J., Welsh, T. N., & Pratt, J. (2019). It is not in the details: Self-related shapes are rapidly classified but their features are not better remembered. *Memory & Cognition*, 47(6), 1145–1157. <https://doi.org/10.3758/s13421-019-00924-6>
- Desebrock, C., & Spence, C. (2021). The Self-Prioritization Effect: Self-referential processing in movement highlights modulation at multiple stages. *Attention, Perception, & Psychophysics*, 83(6), 2656–2674. <https://doi.org/10.3758/s13414-021-02295-0>
- Ghai, S., Forscher, P. S., & Hu, C.-P. (2024). Big-team science does not guarantee generalizability. *Nature Human Behaviour*, 8(6), 1053–1056. <https://doi.org/10.1038/s41562-024-01902-y>
- Ghai, S., Thériault, R., Forscher, P., Shoda, Y., Syed, M., Puthillam, A., Peng, H. C., Basnight-Brown, D., Majid, A., Azevedo, F., & Singh, L. (2025). A manifesto for a globally diverse, equitable, and inclusive open science. *Communications Psychology*, 3(1), 16–24. <https://doi.org/10.1038/s44271-024-00179-1>
- Gold, J. I., & Shadlen, M. N. (2007). The neural basis of decision making. *Annual Review of Neuroscience*, 30(1), 535–574. <https://doi.org/10.1146/annurev.neuro.29.051605.113038>
- Golubickis, M., & Macrae, C. N. (2021). Judging me and you: Task design modulates self-prioritization. *Acta Psychologica*, 218, 103350. <https://doi.org/10.1016/j.actpsy.2021.103350>
- Grossmann, I., & Jowhari, N. (2018). Cognition and the self: Attempt of an independent close replication of the effects of self-construal priming on spatial memory recall. *Journal of Experimental Social Psychology*, 74, 65–73. <https://doi.org/10.1016/j.jesp.2017.08.005>
- Haaf, J. M., Hoffstadt, M., & Lesche, S. (2024). *Attentional control data collection: A resource for efficient data reuse*. PsyArXiv. <https://doi.org/10.31234/osf.io/4evy6>
- Houtkoop, B. L., Chambers, C., Macleod, M., Bishop, D. V. M., Nichols, T. E., &

- Wagenmakers, E.-J. (2018). Data sharing in psychology: A survey on barriers and preconditions. *Advances in Methods and Practices in Psychological Science*, 1(1), 70–85. <https://doi.org/10.1177/2515245917751886>
- Kirk, N. W., & Cunningham, S. J. (2025). Listen to yourself! Prioritization of self-associated and own voice cues. *British Journal of Psychology*, 116(1), 131–148. <https://doi.org/10.1111/bjop.12741>
- Lemaire, P., & Brun, F. (2016). Effects of problem presentation durations on arithmetic strategies: A study in young and older adults. *Journal of Cognitive Psychology*, 28(8), 909–922. <https://doi.org/10.1080/20445911.2016.1215318>
- Lin, Z., Ma, Q., Huang, X., Wu, X., & Zhang, Y. (2023). Pervasive failure to report properties of visual stimuli in experimental research in psychology and neuroscience: Two metascientific studies. *Psychological Bulletin*, 149(7–8), 487–505. <https://doi.org/10.1037/bul0000399>
- Liu, Z., Hu, M., Zheng, Y., Sui, J., & Chuan-Peng, H. (2025). A multiverse assessment of the reliability of the self-matching task as a measurement of the self-prioritization effect. *Behavior Research Methods*, 57(1), 37–56. <https://doi.org/10.3758/s13428-024-02538-6>
- Markus, H. R., & Kitayama, S. (1991). Culture and the self: Implications for cognition, emotion, and motivation. *Psychological Review*, 98(2), 224–253. <https://doi.org/10.1037/0033-295X.98.2.224>
- Milham, M. P., Craddock, R. C., Son, J. J., Fleischmann, M., Clucas, J., Xu, H., Koo, B., Krishnakumar, A., Biswal, B. B., Castellanos, F. X., Colcombe, S., Di Martino, A., Zuo, X.-N., & Klein, A. (2018). Assessment of the impact of shared brain imaging data on the scientific literature. *Nature Communications*, 9(1), 2818. <https://doi.org/10.1038/s41467-018-04976-1>
- Moreau, D., & Gamble, B. (2022). Conducting a meta-analysis in the age of open science: Tools, tips, and practical recommendations. *Psychological Methods*, 27(3), 426–432. <https://doi.org/10.1037/met0000351>
- Munafò, M. R., Nosek, B. A., Bishop, D. V. M., Button, K. S., Chambers, C. D., Percie Du Sert, N., Simonsohn, U., Wagenmakers, E.-J., Ware, J. J., & Ioannidis, J. P. A. (2017). A manifesto for reproducible science. *Nature Human Behaviour*, 1(1), 0021. <https://doi.org/10.1038/s41562-016-0021>
- Poldrack, R. A., Markiewicz, C. J., Appelhoff, S., Ashar, Y. K., Auer, T., Baillet, S., Bansal, S., Beltrachini, L., Benar, C. G., Bertazzoli, G., Bhogawar, S., Blair, R. W., Bortoletto, M., Boudreau, M., Brooks, T. L., Calhoun, V. D., Castelli, F. M., Clement, P., Cohen, A. L., ... Gorgolewski, K. J. (2024). The past, present, and future of the brain imaging data structure (BIDS). *Imaging Neuroscience*, 2, 1–19. https://doi.org/10.1162/imag_a_00103
- Rahnev, D. (2025). A comprehensive assessment of current methods for measuring metacognition. *Nature Communications*, 16(1), 701–719. <https://doi.org/10.1038/s41467-025-56117-0>
- Rahnev, D., Desender, K., Lee, A. L. F., Adler, W. T., Aguilar-Lleyda, D., Akdoğan, B., Arbuzova, P., Atlas, L. Y., Balci, F., Bang, J. W., Bègue, I., Birney, D. P., Brady, T. F., Calder-Travis, J., Chetverikov, A., Clark, T. K., Davranche, K., Denison, R. N., Dildine, T. C., ... Zylberberg, A. (2020). The confidence database. *Nature Human Behaviour*, 4(3), 317–325. <https://doi.org/10.1038/s41562-019-0813-1>
- Ratcliff, R., Smith, P. L., Brown, S. D., & McKoon, G. (2016). Diffusion decision model: Current issues and history. *Trends in Cognitive Sciences*, 20(4), 260–281.

<https://doi.org/10.1016/j.tics.2016.01.007>

Robbins, K., Truong, D., Jones, A., Callanan, I., & Makeig, S. (2022). Building FAIR Functionality: Annotating Events in Time Series Data Using Hierarchical Event Descriptors (HED). *Neuroinformatics*, 20(2), 463–481. <https://doi.org/10.1007/s12021-021-09537-4>

Rodriguez, J. E., & Williams, D. R. (2022). psymetadata: An R package containing open datasets from meta-analyses in psychology. *Journal of OPEN Psychology DATA*, 10(1), 8–13. <https://doi.org/10.5334/jopd.61>

Schäfer, S., Frings, C., & Wentura, D. (2016). About the composition of self-relevance: Conjunctions not features are bound to the self. *Psychonomic Bulletin & Review*, 23(3), 887–892. <https://doi.org/10.3758/s13423-015-0953-x>

Schäfer, S., Wesslein, A.-K., Spence, C., Wentura, D., & Frings, C. (2016). Self-prioritization in vision, audition, and touch. *Experimental Brain Research*, 234(8), 2141–2150. <https://doi.org/10.1007/s00221-016-4616-6>

Sui, J., & Gu, X. (2017). Self as Object: Emerging Trends in Self Research. *Trends in Neurosciences*, 40(11), 643–653. <https://doi.org/10.1016/j.tins.2017.09.002>

Sui, J., He, X., Golubickis, M., Svensson, S. L., & Neil Macrae, C. (2023). Electrophysiological correlates of self-prioritization. *Consciousness and Cognition*, 108, 103475. <https://doi.org/10.1016/j.concog.2023.103475>

Sui, J., He, X., & Humphreys, G. W. (2012). Perceptual effects of social salience: Evidence from self-prioritization effects on perceptual matching. *Journal of Experimental Psychology: Human Perception and Performance*, 38(5), 1105–1117. <https://doi.org/10.1037/a0029792>

Sui, J., & Humphreys, G. W. (2015). The Integrative Self: How Self-Reference Integrates Perception and Memory. *Trends in Cognitive Sciences*, 19(12), 719–728. <https://doi.org/10.1016/j.tics.2015.08.015>

Sui, J., & Rotshtein, P. (2019). Self-prioritization and the attentional systems. *Current Opinion in Psychology, Attention & Perception*, 29, 148–152. <https://doi.org/10.1016/j.copsyc.2019.02.010>

Sui, J., Rotshtein, P., & Humphreys, G. W. (2013). Coupling social attention to the self forms a network for personal significance. *Proceedings of the National Academy of Sciences*, 110(19), 7607–7612. <https://doi.org/10.1073/pnas.1221862110>

Sun, S., Wang, N., Wen, J., & Hu, C.-P. (2023). A dataset of cognitive ontology for neuroimaging studies of self-reference. *China Scientific Data*, 8(3), 175–189. <https://doi.org/10.11922/11-6035.csd.2022.0047.zh>

Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., Da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., ... Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3(1), 160018. <https://doi.org/10.1038/sdata.2016.18>

